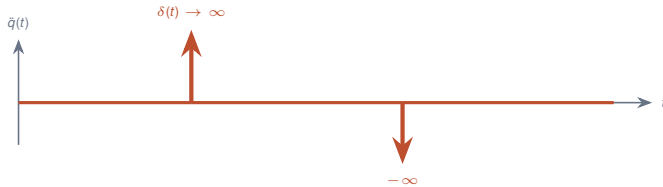
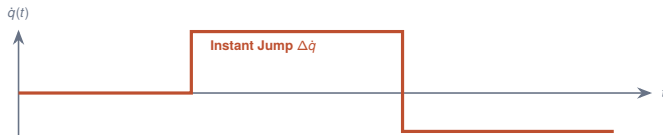


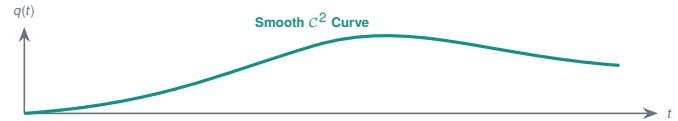
DERIVATIVE CONTINUITY C^2 VS. DISCONTINUOUS WAYPOINT STEPPING

Actuator Protection & Gearbox Shock Mitigation

❌ Raw Chunk Stepping (C^0 Failure → Hardware Shock)



✅ C^2 Quintic Spline ($\|\ddot{q}\| \leq j_{\max}$ Continuous)



⚠️ Hardware Destruction:

- Infinite jerk ($\ddot{q} \rightarrow \infty$) creates high-frequency torque shock.
- Shears cycloidal drive pins & harmonic flexspline teeth.
- Excites mechanical chatter; triggers motor overcurrent trip.

🛡️ Actuator Protection:

- Jerk bounded strictly within $\|\ddot{q}\| \leq j_{\max} = 25 \text{ rad/s}^3$.
- Zero gear backlash impact; smooth torque transmission.
- Motor coil current $I(t)$ remains within continuous thermal envelope.